12.1 Trading Systems

Learning Objective Statements: Trading Systems

Rick Martin, CMT

Learning Objective Statements

- Compare five common trading models
- Outline the go/no-go metrics
- Explain the four ways of generating trading signals
- Analyze the seven steps to a consistently profitable trading system

Building trading models is deceptively simple; however, building trading systems that are consistently profitable is extremely difficult. Seasoned trading system developers will tell you that not even 1% of their ideas ever make it into production, resulting in maybe one new system per year.

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Define Your Trading Objectives

"Trading system objectives" refer to the specific goals that a trader aims to achieve through their trading activities. These objectives typically vary depending on factors such as the trader's risk tolerance, investment horizon, trading style, and overall financial goals. It is important that each objective be measurable. For example, "an average annual profit target of 15%-20%" is a measurable objective, while "strong returns" is not. Some common trading system objectives include:

1. **Profitability:** Generating consistent profits is often the primary objective of any trading system. Traders aim to outperform the market or achieve a certain rate of return on their investments.
2. **Risk management:** Controlling and minimizing risk exposure is crucial for preserving capital and preventing large losses. Trading systems often incorporate risk management techniques such as position sizing, stop-loss orders, and portfolio diversification.
3. **Consistency:** Consistent performance is desired to ensure long-term success. Trading systems aim to achieve reliable results over time, rather than relying on sporadic gains or losses.
4. **Liquidity:** Ensuring that trades can be executed efficiently at fair market prices is essential for maintaining liquidity. Trading systems may focus on markets with high liquidity to facilitate smooth order execution.
5. **Adaptability**: Markets are dynamic and constantly evolving, so trading systems should be adaptable to changing market conditions. This may involve adjusting trading strategies, parameters, or asset allocations as market trends shift.
6. **Robustness**: Trading systems should be robust enough to withstand various market scenarios, including periods of high volatility or unexpected events. Robust trading systems are less likely to experience significant losses due to temporary market disruptions.
7. **Scalability**: Scalable trading systems can accommodate changes in trading volume or capital without sacrificing performance. Scalability is particularly important for institutional traders or trading firms managing large portfolios.
8. **Transparency and accountability:** Maintaining transparency and accountability in trading operations is essential for building trust with investors and regulatory compliance. Trading systems should be well

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Overall, trading system objectives should be aligned with the trader's broader financial goals, risk tolerance, and investment philosophy. Additionally, these objectives may evolve over time as market conditions change or as the trader's priorities shift. Not every model will have the same objectives.

The following objectives might be typical of a trend-following system:

Objective	Commentary
The model is called Triple Trend and is designed to return consistent leveraged results by reducing the number of losing trades by combining three different trend-following calculations while carefully avoiding curve fitting.	Trend-following models generally generate many small losses combined with a few large winners.
Total annualized return 5%-10%	This model is designed to generate modest returns with very low drawdowns.
Maximum drawdown 5%	Used to set leverage, so the lower the better.
Profit factor 1.25-10.0 (Total profit / Total loss)	Less than 1.25 makes the model susceptible to outsized losses, and greater than 10 suggests curve fitting.
Calmar ratio 200% or greater (Annualized returns / Max drawdown)	Ideally 300% or greater, although 200% is acceptable for models with small drawdowns.
Winning percent greater than 50%	This is a trend-following model, so 30%-50% is the norm, but we can expect 50%+ while being careful to not overly fit the results.
(Avg. win / Avg. loss) equal to 1.5 or higher	A normal trend-following model would see this value at 1.5 or higher, reflecting many small losses and a few large winners.

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Describe the Trading Idea

Select Which Markets to Trade

Equities and futures are the most common markets, although bonds, forex, cryptocurrencies, and options are all good alternatives. Most techniques can be applied to any market, so our focus on equities applies to almost any situation.

Five Common Trading Models

There are many high-level models, but we will focus on just the five most common:

1. **A trend-following model** is a type of investment or trading strategy that attempts to capitalize on sustained movements in asset prices by identifying trends in the market. The idea behind this model is to buy assets when their prices are rising and sell them when prices are falling. The strategy does not aim to predict price movements but rather to follow existing trends.
2. A **mean reversion trading model** is based on the idea that prices, returns, or other financial variables tend to revert to their historical average or mean over time. The underlying belief is that when an asset's price deviates too far from its average, it is likely to revert to that average, creating trading opportunities.
3. **Scalping trading** is a high-frequency, short-term trading strategy used in financial markets. The goal of scalping is to profit from small price changes, often in liquid markets like stocks, forex, or cryptocurrencies. Traders who employ this strategy, called scalpers, make numerous trades throughout the day, holding positions for only seconds or minutes.
4. **Pairs trading** is a market-neutral trading strategy that involves buying one asset and simultaneously selling a correlated asset, betting that the price relationship between the two will converge. The idea is that one of the two assets is underpriced or overpriced relative to the other, and as the prices revert to their historical relationship, the trader can profit from the price difference.
5. **Arbitrage trading** is a strategy where traders exploit price differences of the same or similar financial instruments across different markets or exchanges. The core idea is to profit from the price inefficiencies that exist for a brief period. There are several types of arbitrage, but the general process involves buying an asset in one market where the price is lower and

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Strategy Development

Based on your objectives and market analysis, developing a trading strategy that outlines the conditions for entering and exiting trades is the next step. This could involve technical analysis, fundamental analysis, quantitative analysis, or a combination of approaches called fusion analysis. A strategy should be clearly defined and testable.

1. **Entry rules:** Keep it simple. If a rule cannot be explained in plain English, it will likely be difficult to program and test. Keep the number of parameters small; the more parameters that are used, the more likely the results will be overly fit with little predictive value. This is where many traders fail by searching for the Holy Grail, and when their model fails in live testing, they give up.
2. **Exit rules:** Exits have a huge impact on profitability — more than entries, in most cases. There is no place for discretion in a trading system, and yet many traders override their system when a trade is going against them.
3. **Locate the sweet spot:** Most models will have very different returns depending on the bar interval and holding period used in the signal generator. By running the same model with different bar intervals and holding periods, we can usually find the best interval overall, balancing returns and drawdowns. The following table shows an example of a sweet spot analysis. As you may see, the 60-minute bar produces the best overall results, and you may also see that most of the bar intervals have Calmar ratios north of 1 (100%), suggesting the model delivers consistent results across different trading environments.

Stock	Time Interval	Return	Drawdown	Calmar
TSLA	3	13%	7%	1.85
TSLA	5	14%	4%	3.5
TSLA	10	21%	4%	5.25
TSLA	15	22%	6%	3.66
TSLA	20	20%	5%	4
TSLA	30	22%	7%	3.14
TSLA	60	24%	4%	6
TSLA	120	22%	6%	3.66

4. **Entry/exit testing:** Are your returns primarily coming from your entries or your exits? There are many ways to test this, and the easiest is to replace your exits with a fixed bar exit, let's say 5-10 bars, to see how your entries do. And then replace your entries with a random entry to see how your exits do. This not only informs about the system you are developing, but will likely identify entries and exits that you can use in future models.

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objectives, discard the strategy and start over — it will only get worse from here. There are countless risk/return metrics available, and we will focus on those that we find the most helpful:

a. **Net profit** is the ultimate metric, but it can vary substantially depending on the nature of the strategy. It should be viewed in conjunction with drawdowns.

b. **Profit factor** is gross profit divided by gross loss. The generally accepted range is 1.25 to 10. The most reliable strategies will have profit factors of three to five, and greater than 10 strongly suggests curve fitting.

c. **Total number of trades** should be an absolute minimum of 30, with 100 or more as ideal.

d. **Winning percent** depends on the underlying strategy. For example, trend-following models are typically 30% to 50%, while scalping models will be 80% to 90%.

e. **Average net profit per trade** also varies substantially depending on the underlying strategy.

f. **Average winner divided by average loser** also depends on the underlying strategy. For example, trend-following models typically have values greater than one, while mean reversion models have values that are less than one.

g. **Slippage and commissions** are often overlooked, but it is crucial to include reasonable values for both, particularly with actively traded systems.

h. **Maximum drawdown** should be as low as possible. It not only measures the maximum risk during your testing interval, but is also the primary input to position sizing and leverage.

i. **The Calmar ratio** is one of many risk-to-reward ratios, and one of the most widely used by active traders. This is calculated by taking the portfolio's annualized return and dividing it by the maximum drawdown.

Risk Management

The hardest part of risk assessment is being too optimistic. After all, we are building a trading system to make money, and we generally overestimate our tolerance for losses.

1. **Position sizing and leverage:** Concentration and leverage are our friends most of the time, but rapidly become our enemies when a trade — and worse yet, the market — turns against us. A common approach to setting position size relies on the maximum drawdowns from the walk forward test and/or the Monte Carlo test. Assume that the maximum drawdown from the tests is 12% and the maximum acceptable drawdown is 20%.
   Maximum leverage = Target drawdown / Drawdown from tests

returns during normal market conditions, but they can be effective in preventing catastrophic losses, and even when returns are impacted, risk-adjusted returns might improve. Similarly, profit targets rarely improve returns, but like stops, they might improve risk-adjusted returns.

3. **Diversification:** Trading uncorrelated assets usually reduces drawdowns, improving risk-adjusted returns and possibly absolute returns. In addition, trading correlated assets with different strategies or using different time frames is an alternative way to measure diversification.
4. **Just let it trade:** If you have a fully vetted system that has proven to be successful in live trading, it will have losing periods, and it is tempting to override it — do not. The only way it can achieve its long-term returns is to take every trade. The computer has no opinions, and you should not, either.
5. **Go/no-go:** Another chance to bail before we go live.

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Testing and Signal Generation Methods

With so many failures in the development process, it is crucial that failing ideas be weeded out as early as possible, freeing up time for potential winners. The preliminary steps towards success are:

Establish a testing regimen: There are four basic ways to generate trading signals: historical backtesting, out-of-sample, walk forward, and real-time analysis.

1. **Historical backtesting** is the most common approach, and traders love it since it generally produces the best results — smooth equity curves, modest drawdowns, strongest risk-to-reward metrics. And with a little optimization, the results look even better — so much better that they cannot be achieved in real life. Gently optimizing does have a major advantage in eliminating ideas early on. Results will never be better, so use this testing approach to screen out ideas early in the process.
2. **Out-of-sample testing** helps solve the optimization problem by breaking your data into two pieces: in-sample and out-of-sample. Assume you have five years of data, and that you gently optimize the first three years of in-sample data and then use the optimized parameters for the two years of out-of-sample data.
3. **Walk forward analysis** (WFA) is a backtesting technique that involves optimizing a trading strategy on a segment of historical data (the in-sample period), then testing it on a subsequent segment of unseen data (the out-of-sample period). This process is repeated multiple times to simulate real-time trading and to check if the strategy can adapt to changing market conditions.
    a. **More realistic outcome:** WFA produces a more realistic performance assessment than the standard optimized backtesting by more closely simulating real trading conditions. While intraday models often see similar results comparing WFA with optimized backtests, swing models can be quite different, since positions are held days, weeks, or even months.
    b. **Adapts to changing market:** Unlike optimized backtests, which are static by their very nature, WFA adapts to changing market conditions as it rolls forward. This can be seen in Figure 12.1.1 The yellow curve is SPY. The vertical lines are the out-of-sample periods. The blue curve is the walk forward analysis, which was generally

adapted quickly to the market, maintaining a positive return with minimal drawdowns.

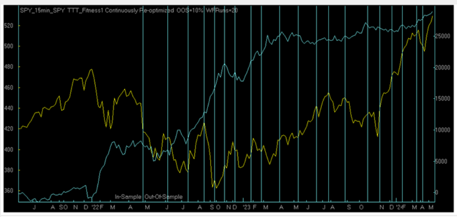

FIGURE 12.1.1 WFA example: yellow curve marks SPY, blue curve marks WFA, blue vertical lines mark out-of-sample periods

i. **Identify effective strategies:** Not only can WFA help identify strategies that are overly fit, but they can also identify those that are generally effective.

ii. **Go/no-go:** This is another decision point; if the WFA does not produce positive results that meet your objectives, discard it and move on.

4. **Real-time analysis** is the most accurate, but it has a major drawback — it can take weeks, months, or even years to have enough trades to produce a valid analysis. Statisticians cannot seem to agree on how many trades you need to generate meaningful statistics. 30-50 seems to be a consensus, but more is always better.

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Monte Carlo

This is the last step in analyzing the trading system before establishing position size to meet risk targets. Monte Carlo analysis is a powerful statistical method used in trading to model the probability of different outcomes in a process that is inherently uncertain, such as the returns of a trading strategy. By simulating many possible future price paths or portfolio outcomes, traders can get a better understanding of the potential risks and rewards involved. Here is how it works in the context of trading:

1. **Purpose of Monte Carlo in trading**
   - **Risk management:** Monte Carlo analysis is often used to estimate the range of potential losses or profits under different market conditions. This helps traders manage risk and optimize their trading strategies.
   - **Probabilistic forecasting:** It allows traders to forecast returns based on statistical distributions, instead of relying on a single point estimate.
   - **Strategy testing:** Traders can simulate the performance of trading strategies under various market scenarios and see how the strategy performs over time.
2. **How Monte Carlo simulations work**
   - **Step 1: Define the model**
     The first step is defining a mathematical model for the trading system that reflects how its price or return might behave. The model often includes assumptions about expected returns, volatility, and correlations with other assets.
   - **Step 2: Random sampling**
     In Monte Carlo analysis, the key is to simulate many different scenarios by randomly sampling from the assumed distributions. These could be normal distributions for asset returns, log-normal distributions for price movements, etc.
   - **Step 3: Run the simulations**
     Thousands or even millions of different potential future price paths are generated based on the random sampling. Each path represents a possible future outcome. Over time, patterns emerge, showing likely and extreme outcomes.
   - **Step 4: Analyze the results**

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  - Worst-case losses
  - Probability of reaching specific targets or thresholds
  - Impact of various levels of volatility

3. **Applications of Monte Carlo in trading**
   Trading strategy evaluation: Before deploying a strategy in the market, traders can run Monte Carlo simulations to understand how the strategy might perform in different market scenarios and across various time frames.

4. **Key assumptions**
   - **Randomness:** Monte Carlo assumes that the future price movements are random and based on probabilistic rules, often following a normal or log-normal distribution.
   - **Stationarity:** Many models assume that the statistical properties of the asset (e.g., volatility and mean return) remain constant over time, although more sophisticated models can account for changing market conditions.

5. **Limitations**
   - **Model dependency:** Monte Carlo results are only as good as the model assumptions. Poor assumptions about asset return distributions, correlations, or volatility can lead to misleading results.
   - **Computational complexity:** Running large-scale simulations can require significant computational power, especially for complex portfolios or derivative structures.
   - **Path dependency:** In certain financial instruments (e.g., path-dependent options), the value depends on the sequence of price movements, adding complexity to the simulation.

Going Live

Once you are satisfied with the performance of your trading system, you can start paper trading live. Monitor the markets closely, adhere to your trading plan, and be prepared to make adjustments as needed based on evolving market dynamics.

1. **Tiptoe in with real money:** When you have at least 50 live simulated trades, go live in a real account with the minimum position size possible.
2. **Increase trade size:** Increase your position size if your returns are as expected after each 30 trades until you reach the maximum position size.
3. **Evaluate and refine:** Continuously monitor the performance of your trading system and make adjustments as necessary. Keep detailed records of your trades and analyze the results to identify areas for improvement.

Remember that developing a successful trading system takes time, patience, and dedication. It is important to approach

Strategy Profiles

Calculating the expectancy for different trading strategies involves understanding the unique characteristics and risk profiles of each strategy. Different trading strategies have unique profiles that affect their expectancy. Traders must understand these profiles to manage risk and optimize their strategies. Regularly calculating and reviewing expectancy helps traders make informed decisions and adapt to changing market conditions.

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The Seven Steps to a Consistently Profitable Trading System

1. **Trading objectives** should be documented up front — it is much more difficult to accept a marginal system if it doesn't meet all our objectives.
2. **Describe the trading idea**, including a general description, markets to test, bar intervals, historical test period, entry and exit rules, and risk management.
3. **Basic testing** relies on the data from the trading idea. This phase uses lightly optimized testing and exhaustive or genetic testing depending on the number of variables.
4. **Walk forward testing** is the core testing approach since it generally adapts to changes in the market, resulting in better results. If this phase does not produce results matching your objectives, discard it and start over, since it will only get worse from here.
5. **Monte Carlo analysis** is a very difficult test, and otherwise good systems can and do fail at this point, either because returns come under pressure or drawdowns expand.
6. **Live trading** in a simulated account or very small positions in a real account are the acid test, and they must meet your objectives. If they do, increase position size a little at a time until you reach your maximum position size.
7. **Add leverage** — or deleverage — based on the maximum drawdown data.

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